

Explanation Report

1. Project Overview

This report explains how the Student Academic Advisor Engine works. The system accepts student facts, checks them against predefined rules, and produces conclusions such as scholarship eligibility, graduation eligibility, probation status, or registration restrictions.

2. Objective

The goal of this project is to show how rule-based reasoning can be used to support academic advising decisions. Instead of manually checking conditions, the system applies rules automatically.

3. Input Facts

- GPA
- Attendance percentage
- Disciplinary status
- Prerequisite completion
- Outstanding fees

4. Knowledge Base

The reasoning logic comes from `knowledge_base.json`. This file stores the academic facts and decision rules used by the engine.

5. Rules Used

1. Scholarship Rule – GPA > 3.5, Attendance > 80%, No disciplinary cases → Eligible for Scholarship
2. Graduation Rule – GPA > 3.0, Prerequisites completed, No outstanding fees → Eligible for Graduation
3. Probation Rule – GPA < 3.0 → Academic Probation
4. Registration Block Rule – Outstanding fees exist → Registration Blocked
5. Dean's List Rule – GPA > 3.5, Attendance > 80% → Dean's List Candidate

6. Reasoning Method

1. Read the student input.
2. Compare each input with the rule conditions.
3. Trigger every rule whose conditions are satisfied.
4. Return the conclusions and explanations.

7. Example Scenario

Sample student input:

- GPA = 3.8
- Attendance = 90%
- No disciplinary cases
- Prerequisites completed
- No outstanding fees

Triggered conclusions:

- Eligible for Scholarship
- Eligible for Graduation
- Dean's List Candidate

8. Why the Conclusions Are Generated

The engine generates these conclusions because the student meets the required conditions for each rule. For example, the scholarship rule is satisfied because the GPA is above 3.5, attendance is above 80%, and there are no disciplinary cases.

9. Result

This report shows that the system works as a simple expert system for academic advising. It is easy

to extend by adding more rules and more student conditions in the future.